

Project Proposal

Fracture Localization in Wrist Radiographs via Healthy Anatomy Reconstruction

Problem and Motivation

Automatic detection of wrist fractures from X-ray images is a common task in biomedical imaging. Most existing approaches rely on supervised classification models to predict whether a fracture is present. However, these models often achieve only moderate performance, especially for subtle fractures, and provide limited interpretability.

To address these limitations, we propose an alternative approach based on modeling healthy anatomy. Instead of directly classifying fractures, we aim to reconstruct a fracture-free version of the wrist and use the difference between the original and reconstructed images to identify abnormal regions.

Core Questions

The project aims to answer the following questions:

1. Can a model learn the appearance of a healthy wrist from X-ray images?
2. Can reconstruction differences highlight fracture regions?
3. How does this approach compare to a standard classification model in terms of performance and interpretability?

Proposed Approach

We propose a reconstruction-based anomaly detection framework structured in several stages:

- Data preparation: Use the GRAZPEDWRI-DX dataset as the main source of wrist X-rays. The MURA wrist subset will be used to reinforce learning of healthy anatomy.
- Healthy reconstruction model: Train a generative model; such as a convolutional autoencoder or GAN-based approach; to learn the distribution of healthy wrist images. Given an input X-ray, the model will generate a reconstructed version without fracture.
- Anomaly detection: Compute a difference map between the original and reconstructed images to identify regions that deviate from normal anatomy.
- Fracture localization: Use the anomaly map to highlight or approximate the location of the fracture (weak segmentation or bounding box).
- Baseline comparison: Train a standard classification model (e.g., ResNet or DenseNet) to detect fractures and compare it with the proposed method.

Data and Feasibility

We will use two public datasets:

- GRAZPEDWRI-DX: A dataset of over 20,000 wrist X-ray images with annotations related to trauma.
- MURA (wrist subset): A large dataset of musculoskeletal radiographs labeled as normal or abnormal.

The project is feasible within one semester due to the use of standard architectures, pretrained models, and a modular pipeline (reconstruction, anomaly detection, and localization).

Evaluation Plan

Evaluation will include:

- Classification metrics (baseline): Accuracy, AUROC, Precision, Recall, F1-score
- Reconstruction quality: MSE, SSIM, and visual inspection
- Localization performance: IoU or Dice (if applicable), and qualitative comparison with annotated fracture regions

Special attention will be given to the interpretability of the results.

Team Contributions

- Javier: data preparation and preprocessing
- Lydia: baseline classification model
- Jorge: reconstruction and anomaly detection

All members will contribute to evaluation, analysis, and the final presentation, while also actively supervising each other's work and collaborating closely throughout the project to ensure consistency, quality, and shared understanding across all stages.